

Exactly Solvable Models

1D : Ising.

2D : QM 有对偶.

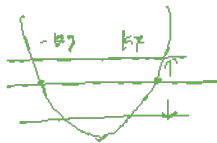
2+1D : tinker toric code.

1+1D : Bethe ansatz, 一堆所积料.

complex plane, 共形场论.

Bosonization, 一堆电子能激发.

$E(k)$



$$\Rightarrow \pm v_F k \pm \int \phi^\dagger i v_F \partial_x \phi dx$$



映射到 1+1D

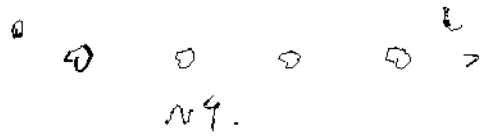
$$\Rightarrow \uparrow \vec{s}_{imp}$$

$\vec{s}_{imp} \cdot \vec{s}$

$$\begin{matrix} E_1 & E_2 & & \\ 0 & 0 & 0 & \dots \end{matrix}$$

一些等价粒子.

§ 1D 刚性小球.



$$\text{相互作用势 } u(r) = \begin{cases} \infty & r \leq D \\ 0 & r > D \end{cases}$$

$$H = H_0 + H_2 \\ = \sum_i \frac{p_i^2}{2m} +$$

配分函数 (正则系综)

$$Z = \text{Tr}(e^{-\beta H}) = \text{Tr}[e^{-\beta(H_0 + H_2)}]$$

$$= Z_0 \cdot Z_2 \quad Z_0 = \left(\frac{2\pi}{\lambda}\right)^N$$

$$= \left(\frac{2\pi}{\lambda}\right)^N \left[\frac{L}{N}\right] \int_{x_1} \dots \int_{x_N} \prod_{i=1}^N dx_i \dots dx_N$$

去重复. ↑ 不接触为1, 接触位置为1

积分空间. $x_1 \in [0, L]$

$$x_2 \in [x_1 + D, L - (N-2)D]$$

⋮

$$x_N \in [x_{N-1} + D, L]$$

$$= \left(\frac{2\pi}{\lambda}\right)^N \frac{L(L-ND)^{N-1}}{N!}$$

$$F = -kT \ln Z = -kT (\ln Z_0 + \ln Z_2) \quad \lambda' = \frac{2\pi}{\lambda}$$

$$= -kT \cdot \left[N \ln \lambda' + \ln L + (N-1) \ln(L-ND) - N(\ln N - 1) \right]$$

$$P = - \left(\frac{\partial F}{\partial L} \right)_{T, N} =$$

$$= \frac{n k T}{1 - n D} \quad n = \frac{N}{L}$$

推广: 一维硬核玻色子模型.

$$\oplus \quad \oplus \quad \oplus \quad \dots$$

产生 / 湮灭 $\hat{b}_x, \hat{b}_x^\dagger$

硬核 (同一个位置 x 不能有两个 Boson) $\hat{b}_x \hat{b}_x^\dagger \equiv 0$

格点. 希尔伯特空间: $|0\rangle, |1\rangle$

$$\hat{b}^\dagger \sim \hat{S}^+, \quad \hat{b} \sim \hat{S}^-$$

$$\text{二体级系统 } \hat{H} = \sum_{ij} \hat{S}_i^x \hat{S}_j^x + \hat{S}_i^y \hat{S}_j^y$$

$$\epsilon_{\vec{k}} = \frac{t}{2} (\cos k_x + \cos k_y)$$



hard-core Boson condensation.



spin-order

{ 1D Ising

(D Lattice.

经典自旋: $\sigma_i = \pm 1$



最近相互作用:

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - \mu_B \sum_i \sigma_i$$

$j = i+1$

$$= -J \sum_i \sigma_i \sigma_{i+1} - \frac{1}{2} \mu_B \sum_i (\sigma_i + \sigma_{i+1})$$

$$Z = \text{Tr}(e^{-\beta H}) = \sum_n \langle n | e^{-\beta H} | n \rangle \quad \text{这里 } |n\rangle \text{ 对应 } \{\sigma_i\} \text{ 为 } (+ - + \dots)$$

$$= \sum \dots \sum e^{\beta \sum_i [J \sigma_i \sigma_{i+1} + \frac{1}{2} \mu_B (\sigma_i + \sigma_{i+1})]}$$

$$= \sum \dots \sum_i e^{\beta [J \sigma_i \sigma_{i+1} + \frac{1}{2} \mu_B (\sigma_i + \sigma_{i+1})]}$$

$$= \sum_{\{\sigma_i\}} \exp \left\{ \beta \left[J \sigma_i \sigma_{i+1} + \frac{1}{2} \mu_B (\sigma_i + \sigma_{i+1}) \right] \right\} \exp \dots$$

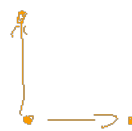
$$\begin{matrix} |+\rangle & & |-\rangle \\ |+\rangle & \begin{pmatrix} e^{\beta(J+\mu_B)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-\mu_B)} \end{pmatrix} & \\ |-\rangle & & \end{matrix} \equiv T_{\sigma\sigma'}$$

$$= \sum_{\{\sigma_i\}} \underbrace{T_{\sigma_1 \sigma_2} T_{\sigma_2 \sigma_3} \dots T_{\sigma_n \sigma_1}}_{\text{trace}}$$

$$= \text{Tr}(T^n)$$



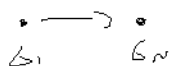
2D:



图像: $\sigma_1 \rightarrow \sigma_2 \rightarrow \sigma_3 \dots$

Tensor Network Renormalization Group

↓



$$\text{对角化: } T \rightarrow T' = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$[e^{\beta(J+\mu_B)} - \lambda] [e^{\beta(J-\mu_B)} - \lambda] = e^{-2\beta J}$$

$$= \lambda - 2e^{\beta} \cosh(\beta \mu B) + 2 \sinh(2\beta J) = 0 \quad \left(\frac{e^{2\beta J} - e^{-2\beta J}}{2} \right)$$

$$\lambda_1, \lambda_2 = e^{\beta J} \left[\cosh(\beta \mu B) \pm \sqrt{\cosh^2(\beta \mu B) - 2 \sinh(2\beta J)} e^{-2\beta J} \right]$$

$$= e^{\beta J} \left[\cosh(\beta \mu B) \pm \sqrt{\sinh^2(\beta \mu B) + e^{-4\beta J}} \right]$$

$$Z = \text{Tr}(T^N) = \lambda_1^N + \lambda_2^N$$

热力学极限 ($N \rightarrow \infty$) 下:

$$\lambda_1^N \gg \lambda_2^N, \quad Z \approx \lambda_1^N$$

$$F = -kT \ln Z$$

$$= -NkT \ln \left\{ e^{\beta J} \left[\cosh(\beta \mu B) + \sqrt{\sinh^2(\beta \mu B) + e^{-4\beta J}} \right] \right\}$$

$$m = - \frac{\partial F}{\partial B}$$

$$= NkT \frac{\partial}{\partial B} \ln \left\{ e^{\beta J} \left[\cosh(\beta \mu B) + \sqrt{\sinh^2(\beta \mu B) + e^{-4\beta J}} \right] \right\}$$

$$= NkT \frac{\partial}{\partial B} \ln \left\{ \cosh(\beta \mu B) + \sqrt{\sinh^2(\beta \mu B) + e^{-4\beta J}} \right\}^{1/2}$$

$$= NkT \frac{1 + \frac{1}{2} \left[\sinh^2(\beta \mu B) + e^{-4\beta J} \right]^{-1/2} 2 \cosh(\beta \mu B)}{\cosh(\beta \mu B) + \sqrt{\sinh^2(\beta \mu B) + e^{-4\beta J}}^{1/2}} \quad \beta \mu \sinh(\beta \mu B)$$

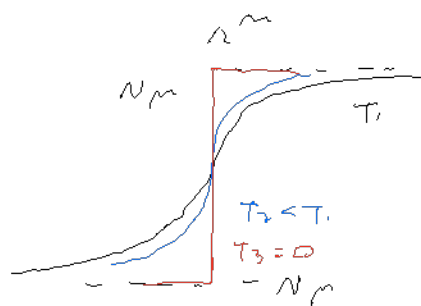
$$= NkT \frac{\beta B \sinh(\beta \mu B)}{\left[\sinh^2(\beta \mu B) + e^{-4\beta J} \right]^{1/2}}$$

$$= \frac{NB \sinh(\beta \mu B)}{\left[\sinh^2(\beta \mu B) + e^{-4\beta J} \right]^{1/2}}$$

$T \neq 0, B \rightarrow 0: M \rightarrow 0 \rightarrow$ 非零温下, 没有长程序,

因为涨落: Mermin-Wagner Theorem - 维系统中不能有自发破缺

$T \rightarrow 0, (B \rightarrow \infty), B \neq 0: M \rightarrow N \mu B \rightarrow$ 零温下产生长程序. 连续对称性.



$B > 0$

$$S_i \rightarrow -S_{i+1}$$

产生 Ising Order, 破坏了 Hamilton 中 $S_i S_{i+1}$ 的 左右对称性. (与 $S_z \sim$)

§ 求解关联函数

$$\langle G_i, G_{i+1} \rangle \sim \frac{\partial^2 \langle J \rangle}{\partial J^2}$$

$$\langle G_i, G_j \rangle \rightarrow \text{中间插入 } G_{i+1} \dots G_{j-1}$$

{ 2D Ising Model

$$H = - \sum_{\langle i,j \rangle} J_{ij} S_i S_j$$

1. 高温展开

$$Z = \text{Tr}(e^{-\beta H}) = \bar{Z}$$

$$= \sum_{\{S_i\}} \cosh^N K \prod_{\langle i,j \rangle} (1 + S_i S_j) \quad K = \beta J$$

$$= 2^N \cosh^N K \sum_{n=0}^N \text{tanh}^N K n(r)$$

N : site num

N : bond num

$n(r)$: closed connection num at r -order



$E = 0$



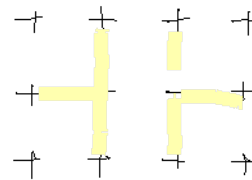
$E \neq 0$



2. 低温展开

$Z =$

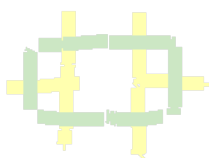
$$= e^{KN} \sum_{n=0}^{\infty} m(r) e^{-2KN}$$



$m(r)$: antibond num at r -order,

3. Dual lattice. (high $\sim$ low)

$$\begin{cases} n(r) = n_{\text{dual}}(r) \\ m(r) = n_{\text{dual}}(r) \end{cases}$$



define k^* : $\tanh k^* = e^{-2k}$

$$\hookrightarrow \sinh(2k^*) \sinh(2k) = 1$$

$$\textcircled{1} \quad Z_{\text{low}} = e^{\frac{1}{2}N} \sum_r m(r) e^{-2kr}$$

Duality

$$= Z_{\text{high}}^* = (\cosh k^*)^{\frac{1}{2}N} \sum_r n_D(r) V^{*r} \quad V^* = \tanh k^* = e^{-2k}$$

$\sum$

$$N_D = N, N_0 = N = 2N$$

$$Z_D(k^*) = Z^{N_D} \cosh^{N_0}(k^*) \sum_r n_D(r) V^{*r} \quad (\text{high } T, \text{ at dual lattice})$$

$$\Rightarrow Z_{\text{low}} = Z_{\text{high}}^* = Z^N \cosh^{2N}(k^*) \tanh^N(k^*) Z_D(k^*)$$

$$= \sinh(2k^*)^N Z_D(k^*) = \frac{\sinh(\frac{N}{2} 2k^*)}{\sinh^{\frac{N}{2}}(2k)} Z_D(k^*)$$

$$\Rightarrow \sinh(2k) Z(T) = \sinh^{-\frac{N}{2}}(2k^*) Z(T^*)$$

相变点: $\sinh(2k_c) = 1$

$$k_c = \frac{1}{2} \operatorname{arcsinh} 1 = 0.441$$

Different lattice?



Different dimension?



Q. self duality of 1D transverse Ising model

$$H = - \sum_{\langle i,j \rangle} J \sigma_i^z \sigma_j^z - \mu B \sum \sigma_i^x$$

$$\begin{cases} \sigma_i^x \rightarrow \sigma_i^z \sigma_{i+1}^z \\ \sigma_i^z \sigma_{i+1}^z \rightarrow \sigma_i^x \end{cases}$$